Assignment No. 9

9. A transport management system maintains common information for all vehicles while storing additional details specific to
different vehicle categories. Design an application that organizes these records efficiently using inheritance.

```
#include <iostream>
#include <string>

using namespace std;

// Base class
class Vehicle {
protected:
    string registrationNo;
    string brand;

public:
    // Constructor using initializer list and const reference
    Vehicle(const string& reg, const string& b)
        : registrationNo(reg), brand(b) {}

    // Virtual destructor for safe inheritance
    virtual ~Vehicle() = default;

    void displayCommon() const {
        cout << "Registration No: " << registrationNo << "\n";
        cout << "Brand: " << brand << "\n";
    }
};

// Derived class: Car
class Car : public Vehicle {
private:
    int seats;

public:
    Car(const string& reg, const string& b, int s)
        : Vehicle(reg, b), seats(s) {}

    void display() const {
        cout << "Car Details\n";
        displayCommon();
        cout << "Seating Capacity: " << seats << "\n";
```

```
        cout << "------------------------\n";
    }
};

// Derived class: Truck
class Truck : public Vehicle {
private:
    double loadCapacity;

public:
    Truck(const string& reg, const string& b, double load)
        : Vehicle(reg, b), loadCapacity(load) {}

    void display() const {
        cout << "Truck Details\n";
        displayCommon();
        cout << "Load Capacity: " << loadCapacity << " tons\n";
        cout << "------------------------\n";
    }
};

int main() {
    Car car("MH12AB1234", "Toyota", 5);
    Truck truck("MH14CD5678", "Tata", 12.5);

    car.display();
    truck.display();

    return 0;
}
```

main.cpp

Run ▶

```
        : Vehicle(reg, b),
loadCapacity(load) {}

    void display() const {
        cout << "Truck Details\n";
        displayCommon();
        cout << "Load Capacity: " <<
loadCapacity << " tons\n";
        cout << "------------------------\n";
    }
};

int main() {
    Car car("MH12AB1234", "Toyota", 5);
    Truck truck("MH14CD5678", "Tata", 12.5);

    car.display();
    truck.display();

    return 0;
}
```

Output

```
Car Details
Registration No: MH12AB1234
Brand: Toyota
Seating Capacity: 5
------------------------
Truck Details
Registration No: MH14CD5678
Brand: Tata
Load Capacity: 12.5 tons
------------------------
```